TRESATA TAKE-HOME ASSIGNMENT · ENGINEERING & ARCHITECTURE BRIEF

Neuro-Symbolic Tax Audit Triage for Indian Section 43B(h) MSME Disallowances

Jurisdiction: India (Income Tax Act, 1961 read with MSMED Act, 2006) · Target Workflow: Statutory Form 3CD Clause 22 Certification

1. Area Explored & Domain Insights That Surprised Me

I explored **Statutory Corporate Tax Auditing in India (Form 3CD under Section 44AB of the Income Tax Act, 1961)**, specifically examining how tax auditors verify trade-payable deductibility for Micro and Small Enterprises (MSEs) under **Section 43B(h)**. Introduced by the Finance Act, 2023, Section 43B(h) mandates that any sum payable by an assessee to a Micro or Small Enterprise beyond the time limit specified in Section 15 of the Micro, Small and Medium Enterprises Development (MSMED) Act, 2006 is deductible only in the financial year in which it is actually paid.

At first glance, Section 43B(h) reads like a trivial calendar check that an ERP SQL query should solve: if a buyer does not pay a Micro or Small vendor within **15 days** (without a written agreement) or up to **45 days** (with a written agreement), the unpaid year-end expense is disallowed and taxed at the corporate rate (~25.17%). Reading actual ICAI technical guidance, Ministry of MSME Office Memoranda, and real vendor audit working papers revealed three non-obvious domain traps that make pure ERP automation impossible:

1. **The First Proviso Exclusion Trap:** Under almost every other clause of Section 43B (such as GST, provident fund, or employee bonuses), a corporate taxpayer can still claim the deduction in Financial Year (FY) 2025–26 as long as it pays the liability before filing its Income Tax Return (typically October 31 of the following year under Section 139(1)). However, Parliament **explicitly excluded Clause (h) from the First Proviso**. If an overdue MSE invoice crosses March 31 unpaid, the deduction is lost for that fiscal year even if settled on April 2.
2. **The Statutory Clock Starts on Acceptance or Defect Resolution, Not Invoice Date:** Under Section 15 and Section 2(b) of the MSMED Act, the 15- or 45-day window ("Appointed Day") is anchored to actual delivery of goods or rendering of services. Crucially, if the buyer raises a **written objection regarding physical defects or quality within 15 days of delivery**, the statutory clock pauses and restarts only on the date the vendor removes the defect. Those facts never exist in ERP ledger columns; they live in scanned Delivery Challans, Goods Receipt Notes (GRNs), and email threads.
3. **Udyam Registration Does Not Automatically Mean Eligibility:** Section 43B(h) applies strictly to *Micro* and *Small* enterprises engaged in **Manufacturing or Services**. *Medium* enterprises are excluded. Furthermore, under Ministry of MSME Office Memorandum No. 5/2(2)/2021-E/P&G/Policy (dated July 2, 2021), **Wholesale and Retail Traders** (NIC codes 45, 46, and 47) are permitted to obtain Udyam Registration certificates strictly for Priority Sector Lending—not for Section 15 delayed-payment protection. A binary ERP "MSME = Yes" flag routinely misclassifies traders and dual-activity vendors.

2. Problem Statement & Scope Boundaries

Dimension	Specification
Who	Chartered Accountants (CAs), corporate tax managers, and statutory audit teams preparing and certifying **Form 3CD (Clause 22)** for mid-to-large Indian enterprises.
What Task	Cross-referencing five unstructured and semi-structured artifacts for every open year-end vendor payable—(1) Udyam Registration Certificate PDF, (2) Purchase Order (PO) or Master Service Agreement (MSA), (3) Tax Invoice, (4) Delivery Challan / GRN, and (5) Quality-Dispute Email Correspondence—to compute exact statutory due dates and tax disallowances.
How Often	Annually during statutory tax audit season (April–September) for final sign-off, plus quarterly (June, September, December, March) for corporate advance-tax provisioning.
What It Costs Today	A mid-market manufacturing company with 3,000 vendors and 800 open March 31 trade payables expends **120–200 auditor-hours** manually reviewing PDFs and reconciling spreadsheets. Missing a disallowance (false negative) triggers a **200% penalty for misreporting of income under Section 270A** and exposes the signing CA to NFRA/ICAI disciplinary action. Over-disallowing an exempt Trader or disputed invoice (false positive) unnecessarily locks up 25.17% of the invoice principal in immediate cash tax outflow for an entire year.
Out of Scope	Automated ERP journal posting or payment execution; auditing non-MSME limbs of Section 43B (clauses (a)–(g) such as statutory taxes, leave encashment, or bank interest); and representing taxpayers in appellate litigation.

3. Solution Design & Architectural Reasoning

Why a Simpler Non-AI Approach Fails — And Why a Single LLM Call Also Fails

A pure SQL or regex rules engine fails because critical audit facts are locked in unstructured prose: regex cannot read a five-message email chain between a plant engineer and a supplier to distinguish a *valid written quality objection raised within 15 days of delivery* (which legally resets the statutory clock) from a commercial request for a credit extension (which does not). Nor can regex distinguish whether "Net 30 Days" was bilaterally agreed in a countersigned PO versus unilaterally printed by a supplier in invoice footer text (which Indian courts treat as "no written agreement," defaulting the statutory cap to 15 days), or determine whether an engineering invoice represents P&L Revenue Expenditure (OpEx, subject to Section 43B(h)) versus Capital Expenditure (CapEx, outside Section 43B(h)).

Conversely, a single-prompt LLM wrapper ("Read these PDFs and output the Section 43B(h) disallowance") is unsafe for statutory audit. LLMs are probabilistic next-token predictors: they hallucinate day counts across February, conflate the March 31 fiscal cutoff with the October 31 ITR filing date, and cannot guarantee identical arithmetic across runs. Our design therefore enforces a strict **Neuro-Symbolic Separation of Concerns**: the LLM operates exclusively as a *probabilistic semantic sensor* that populates a grounded JSON schema (`EvidencePacket`), while a *deterministic Python calendar and rules engine* executes 100% of the statutory logic, date arithmetic, and rupee calculations.

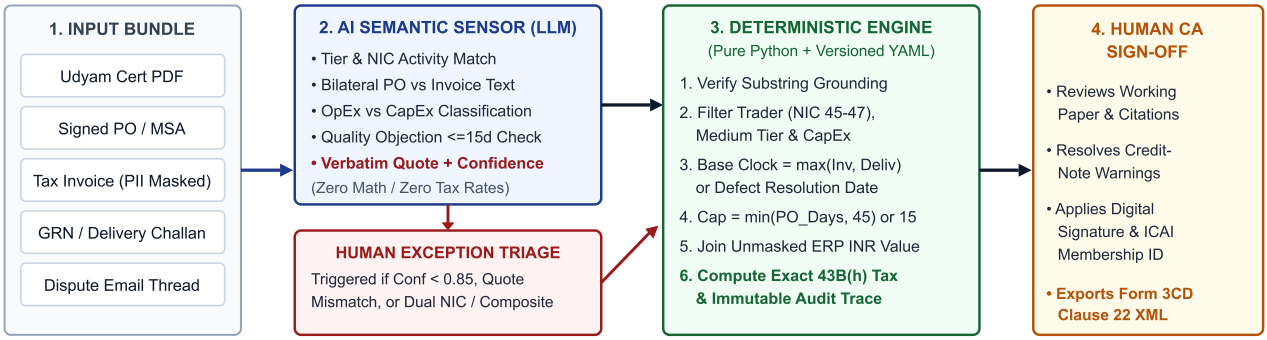


Figure 1: Neuro-Symbolic Audit Pipeline demarcating Probabilistic AI Extraction (blue), Deterministic Python Statutory Evaluation (green), and Mandatory Human Chartered Accountant Checkpoints (red/amber).

Where the Knowledge Lives, Traceability & Handling Rule Expiration

A core architectural trap in financial AI is baking mutable tax law into model prompts or fine-tuned weights. In our system, knowledge is partitioned across four distinct layers with explicit expiration rules:

Knowledge Layer	Where It Lives	Audit Traceability & How Expiration Is Handled
1. Document Semantics (reading layouts, dispute tone, CapEx vs. OpEx)	LLM Weights + Strict JSON Extraction Schema (`EvidencePacket`)	Every extracted field requires a verbatim `evidence_quote` verified deterministically via normalized substring matching against the source OCR text. Swapping the underlying LLM never alters statutory logic.
2. Statutory Tax Law & Thresholds (15/45-day caps, NIC exclusions, MSME limits)	Versioned Configuration File (`rules_AY2026_27.yaml`) + Pure Python Engine	When a Union Budget revises MSME investment/turnover thresholds (as occurred in Budget 2025) or amends Section 43B(h), engineers publish a new date-stamped rule file keyed by Assessment Year. Prior-year audits remain 100% reproducible.
3. Source Evidence Artifacts (Invoices, POs, GRNs, Emails)	Content-Addressed Document Store (SHA-256 hashes + page/character offsets)	Every disallowance line item in the generated Form 3CD working paper links directly to the highlighted page and character span of the source PDF so an Income Tax Assessing Officer can verify it years later.
4. Statutory Accountability	Licensed Human Chartered Accountant (ICAI Member)	Under Section 44AB of the Income Tax Act, only a practicing CA can certify Form 3CD. The pipeline cannot export a final schedule without cryptographic sign-off logging the CA's ICAI Membership Number and UDIN.

4. Alternatives Rejected & Weakest Point of the Design

What We Considered and Rejected

- **Single-Pass LLM Calculation (Prompt-Only End-to-End):** Rejected because LLMs fail on calendar edge cases (such as leap-year February day counts or crossing the March 31 fiscal boundary) and require sending raw invoice pricing to external APIs. Separating extraction from calculation gives 100% arithmetic reliability while keeping the LLM blind to confidential rupee amounts.
- **Fine-Tuning a Domain LLM on Indian Tax Case Law:** Rejected because Indian tax rules change every February 1 in the Union Budget. Embedding statutory thresholds in neural weights creates silent obsolescence and makes it impossible to prove to an auditor which version of the law was applied.
- **Multi-Agent Debate (Taxpayer Agent vs. Tax Inspector Agent):** Rejected because statutory compliance triage is governed by bright-line statutory rules, not subjective negotiation. Multi-agent debate multiplies token cost and latency by 5–6× while muddying the audit trail between factual extraction and statutory math.

Weakest Point of This Design & What Would Break It

The single weakest point of this architecture is **unrecorded offline dispute evidence (the "WhatsApp / Gate-Pass Gap")**. In Indian manufacturing, buyers frequently notify suppliers of defective batches via WhatsApp voice/text messages, telephone calls, or handwritten remarks on physical carbon-copy gate passes (*kaccha challans*) rather than formal emails. If the audit bundle contains an invoice dated January 10 and an April 15 payment, but omits the WhatsApp defect notice sent on January 18 and resolved on March 25, Stage 1 sees no written objection and Stage 2 deterministically flags a **false-positive disallowance**.

Architectural Mitigation: We added a deterministic heuristic in Stage 2 (`Gate 6: OFFLINE_DISPUTE_PROXY`). Whenever an overdue invoice is accompanied by an ERP credit note, debit adjustment, or short payment—the primary accounting footprint of a settled quality dispute—the engine halts automatic disallowance and routes the record to the human CA with a targeted prompt: *"Credit note detected on late invoice—check with procurement for offline gate-pass or WhatsApp quality dispute records before disallowing."*

5. Comprehensive Risk Matrix & Future Considerations

Risk Category	Concrete Failure Scenario	Design Mitigation
Wrong Answers (Hallucination / OCR Error)	LLM misreads an ambiguous date (`03/02/2026` as March 2 instead of Feb 3) or invents a PO clause.	Programmatic substring check enforces that `evidence_quote` exists verbatim in the source text; extraction confidence below `0.85` or ambiguous date formats route to the Human Exception Queue.

Risk Category	Concrete Failure Scenario	Design Mitigation
Outdated Statutory Rules	Government amends MSME turnover ceilings or adds new NIC exemptions mid-year.	Zero statutory thresholds live in prompts. All rules reside in immutable, version-tagged YAML/JSON files (`rule_version: IN_IT_ACT_43BH_AY2026_27_v1.2`) stamped onto every output row.
Confidential Data Exposure	Leaking sensitive vendor PANs, bank account numbers, or proprietary unit pricing to external LLM providers.	Pre-LLM regex scrubber masks PANs, GSTINs, bank details, and exact rupee totals before Stage 1 extraction. Stage 2 joins exact INR amounts directly from the local ERP ledger inside Python.
Legal Liability & Tax Advice	Unqualified software output treated as a binding tax opinion, violating ICAI and Income Tax regulations.	System outputs *Draft Audit Working Papers* only. Final Form 3CD Clause 22 export requires explicit human CA review and digital sign-off.

What I Would Consider Before Spending More Time on This Problem: First, before building custom OCR pipelines for scanned Udyam PDFs, I would evaluate direct API integration with the government's **Udyam API Setu / GSTN portal** to pull structured vendor NIC codes and enterprise tiers directly via GSTIN, restricting LLM usage strictly to contracts, invoices, and dispute correspondence. Second, I would benchmark prototype accuracy across 12 synthetic test cases—where our prototype achieved 10/12 exact passes and surfaced two instructive Stage 1 failure modes: (1) numeric `DD/MM` vs. `MM/DD` date ambiguity on scanned challans, and (2) dual-NIC Udyam certificates where a manufacturer also trades imported spares under a generic invoice description. Both are solvable via deterministic pre-validation gates in v2.

APPENDIX: WHAT I ADDED BEYOND THE AI (1 PAGE)

1. Where AI Helped & Three Key Prompts Used

I used an LLM as a rapid domain-research partner and synthetic test-case generator across three areas: (a) mapping statutory intersections between the Income Tax Act, 1961 and the MSMED Act, 2006; (b) stress-testing failure modes in Indian manufacturing procurement; and (c) generating 12 realistic procurement document bundles for prototype benchmarking.

- **Prompt 1 (Statutory Boundary Discovery):** *"List the exact statutory conflicts and edge cases between Section 43B(h) of the Indian Income Tax Act, 1961 and Sections 2(b), 15, and 16 of the MSMED Act, 2006. Specifically address: (a) how the First Proviso to Section 43B interacts with clause (h), (b) how written quality objections within 15 days alter the 'appointed day', and (c) how composite contracts or traders with Udyam registration are treated."*
- **Prompt 2 (Adversarial Practitioner Critique):** *"Act as a skeptical senior partner at an Indian Big-4 tax audit firm reviewing an AI pipeline that extracts dates and vendor categories from invoices and Udyam PDFs to compute Section 43B(h) disallowances. What real-world document messiness in Indian manufacturing procurement will break this system in production?"*

- **Prompt 3 (Synthetic Benchmark Generation):** *"Generate 12 synthetic test cases representing Indian corporate vendor payables across the March 31 fiscal year-end, including Net-60 PO caps, unilateral invoice footers, timely vs. belated quality objections, Udyam-registered wholesale traders, CapEx machinery invoices, and adversarial OCR traps."*

2. What I Rejected or Corrected from the AI (And How I Knew)

1. **Corrected Fatal Legal Error on the Section 139(1) ITR Filing Due Date:** In initial brainstorming, the LLM stated that an overdue MSME invoice from February 2026 paid in May 2026 is still deductible in FY 2025–26 because it was paid before the October 31 tax return filing deadline under the First Proviso to Section 43B. I verified the actual Finance Act, 2023 statutory text, which explicitly amended the First Proviso to read: *"nothing contained in this proviso shall apply to any sum referred to in clause (h)."* Trusting the LLM's default legal summary would have invalidated every single post-March 31 calculation.
2. **Rejected End-to-End Prompt Calculation:** The AI initially proposed a single chain-of-thought prompt that extracts invoice dates, counts elapsed days, applies the 45-day cap, and outputs the disallowed rupee amount. I rejected this because auditors cannot sign off on probabilistic arithmetic, and sending raw rupee balances to an external API creates unnecessary confidentiality risk when the LLM only needs to extract semantic facts.
3. **Corrected Naive Udyam Certificate Eligibility:** The AI initially suggested checking only whether `enterprise_tier in ["Micro", "Small"]` on the Udyam PDF. By cross-checking the July 2, 2021 Ministry of MSME Office Memorandum and ICAI guidance notes, I identified that Wholesale and Retail Traders (NIC 45, 46, 47) are permitted to register on the Udyam portal solely for Priority Sector Lending and must be deterministically excluded from Section 43B(h) disallowances.

3. What Came From Me (Original Decisions, Insights & Trade-Offs)

- **Amount-Blind Semantic Extraction (Privacy + Precision Trade-Off):** I architected the pipeline so the LLM never receives the invoice's rupee amount, vendor PAN, or bank account number. The LLM extracts only temporal and semantic attributes (dates, bilateral vs. unilateral terms, OpEx vs. CapEx, quality dispute validity), while Stage 2 joins the exact INR balance directly from the local ERP ledger.
- **Deterministic Substring Grounding Gate:** Rather than trusting the LLM's self-reported confidence score alone, I implemented a programmatic check (`verify_grounding`) requiring the extracted `evidence_quote` to appear as a normalized verbatim substring inside the source document text before Stage 2 is allowed to run.
- **The Credit-Note Proxy for Offline Disputes:** Recognizing that factory floor quality disputes often occur on WhatsApp or physical gate passes rather than formal emails, I designed the Stage 2 heuristic that uses ERP credit notes or short payments on late invoices as a deterministic trigger to halt automatic disallowance and request offline evidence from the human auditor.

4. ONE THING I WILL STAND BEHIND LIVE WITHOUT AI

I can derive and defend the entire **deterministic statutory state machine for Section 43B(h) and Section 15 of the MSMED Act** live on a whiteboard with zero tools: (1) why the base clock $\mathbf{T_0}$ equals `max(delivery_date, invoice_date)` unless a written defect objection is raised within `delivery_date + 15 days`, which shifts $\mathbf{T_0}$ to `defect_resolution_date`; (2) why the statutory credit window $\mathbf{L}$ equals `min(agreed_PO_days, 45)` for bilateral written agreements and `15` for unilateral invoice terms or no agreement; and (3) why a disallowance triggers if and only if the vendor is a Micro/Small Manufacturer or Service provider, the expense is current-year P&L OpEx, the balance remains unpaid on March 31, and `actual_payment_date > T_0 + L`.